Biochemistry Cume
April 1, 2005

A new bacteria has been discovered living in an extreme environment which has an electron transfer chain that transfers electrons from NO_2^- to O_2 and couples the energy to ATP production. ($NO_3^- + 2\ H^+ + 2\ e^- \rightarrow NO_2^- + H_2O$; $E^{o'} = +0.42$ V). ($1/2\ O_2 + 2\ H^+ + 2\ e^- \rightarrow H_20$; $E^{o'} = +0.82$). Five redox proteins are involved in the electron transfer chain, A, B, C, D, E, and the bacteria have an ATP synthase with the same properties as the one in normal heart mitochondria. Three inhibitors of the respiratory chain have been discovered, I1, I2, and I3. The percent reduction of each redox protein was determined in state 3 (in which maximal coupled electron transfer rates occur), in state 4 (which is inhibited by lack of ADP), or in the presence of inhibitors I1, I2, or I3:

Percent Reduction of Redox Proteins in Physiological States 3 and 4, or in presense of inhibitors I1, I2, or I3 and excess ADP and NO_2^-.

	A	B	C	D	E
State 3	70%	25%	60%	30%	65%
State 4	100%	0%	100%	0%	100%
I1	100%	0%	0%	0%	0%
I2	100%	0%	0%	0%	100%
I3	100%	0%	100%	100%	100%

Questions:

a) (40) Calculate the total exergonic energy $\Delta G^{o'}$ released during electron transfer from 1 mole of NO_2^- to O_2. Calculate the maximum number of moles of ATP that could be synthesized per mole of NO_2^- oxidized, assuming that $[ATP] = [P_i] = 5 \times 10^{-3}$ M, $[ADP] = 5 \times 10^{-4}$ M, and that 100% of the ΔG of electron transfer is converted to ΔG for ATP synthesis.

b) (60) Determine the order of electron transport from NO_2^- to O_2 and the site(s) at which electron transfer is coupled to proton pumping. Also indicate the points at which inhibitors I1, I2, and I3 inhibit. Draw a model for electron transport, proton pumping, and ATP production, giving all stoichiometries. Explain.

Possibly useful formulas:

$\Delta G^{o'} = -RT\ln K_{eq}$; $\Delta G^o = -nF\Delta E_o$
$\Delta G = \Delta G^o + RT\ln\{\text{products/reactants}\} + nF\Delta V$
F = Faraday's constant = 23 kcal/volt; $R = 1.987 \text{cal mol}^{-1} \text{deg}^{-1}$
ΔG^o(ATP hydrolysis) = -7.3 kcal/mol